6pm Guest Speaker

Title: **Introduction to Wireless and Wireline Transceiver**

Short Bio: **Yu, Rui** (IEEE Senior Member) received the Ph.D. degree in ECE Dept., National University of Singapore. He is currently a connectivity chipset design director with **Huawei Technologies**. His research interests are in the area of RF and mixed-signal IC design for connectivity wireless and wireline applications

* students looking for jobs should make use of this opportunity

Followed by pre-Exam review, Q & A

https://www.linkedin.com/in/rui-yu-56617215/
https://en.wikipedia.org/wiki/HiSilicon

Huawei Singapore research center

standard → system requirement –. architecture → perblock spec —> circuit design (18 -24 month)

sensitive of rx typically dominated by NF
Psen = -173.8 + NF db+ 10logBW + SNRmin

- SNR min forms standard
- -173.8 from 50ohm

analog circuit design not good enough for SNR for like wifi74096 QAM

- analog max i s30dB snr
  - due to matching
- for max production is tough
  - so must need calibration
    - thermal noise cannot never be calibrated, some can be canceled
    - calibration can do in
      - run time
      - idle or startup
      - background not stable enough of practical product
    - calibration is using feedback
      - some sense line, this is none ideal so it limits the cal

huawei rf is in shanghai

ISSCC

- near link is a chinese standard, set by huawei
  - to fight against bluetooth

PA will affect VCO, this is called pulling

- they do pulling cancellation
- everything RF
  https://www.everythingrf.com › Technical Articles
  What is Frequency Pulling?
  8 Apr 2021 — Frequency **pulling** is a variation in the frequency of the **VCO**/oscillator due to a change in the load impedance. Read more

SC PA

- like a RF DAC

PLL and rx divider are separate, but the link is simplex why not just turn off one

for the digital sampling 32MHz, why not just be faster than 80 MHz then can avoid

- its good enf?

How is BER and PER measured, with an anechoic chamber? wired?

- standard define the equipment to measure
  - like a loop
- from instrument manufacturer

wireline channel is time varying

- need adaption loop